# Reddit as a Linguistic Ecosystem: Mapping Community Identity Through NLP

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## Abstract

This project investigates Reddit as a linguistic ecosystem - a platform where thousands of topic-specific communities, called subreddits, have evolved distinct vocabularies, tones, and discourse patterns despite sharing the same digital infrastructure. The central question driving this research is: can we quantify how differently communities talk, and what does this reveal about their collective identity?

To answer this, data was collected from 10 carefully selected subreddits spanning work culture, mental health, financial discussion, education, and lifestyle communities, using Reddit's public JSON API. A total of 1,500 posts and approximately 30,000 comments were collected, cleaned, and analysed using a combination of classical machine learning and modern natural language processing techniques.

The pipeline consists of four main components: exploratory data analysis revealing lexical and temporal patterns across communities; a subreddit classifier built using TF-IDF vectorisation and Logistic Regression achieving 64.13% accuracy across 10 classes; dynamic topic modelling using BERTopic discovering 18 distinct thematic clusters; and cross-community word embedding comparison using Word2Vec, revealing how identical words carry fundamentally different semantic associations across subreddits. Results were deployed as a fullstack web application with a React frontend and FastAPI backend, enabling interactive exploration of the findings.

## 1 Introduction

Online communities have become one of the most significant sites of language evolution in the modern era. Platforms like Reddit host millions of users across thousands of topic-specific communities, each developing its own shared vocabulary, rhetorical norms, and linguistic identity. A post from r/antiwork reads nothing like one from r/Entrepreneur, even when both discuss the concept of 'work'. A comment in r/depression carries entirely different semantic weight than one in r/AskReddit, even when using the same words. These differences are not accidental - they are the product of community culture, shared experience, and the reinforcement loops of moderation and upvoting.

This project treats Reddit as a linguistic ecosystem and applies natural language processing techniques to study how 10 distinct communities use language differently. The term 'ecosystem' is deliberate - just as biological ecosystems are defined by the relationships and interdependencies between species, linguistic ecosystems are defined by the relationships between communities, their vocabularies, and their discourse patterns. Some communities overlap heavily in vocabulary; others are so distinct that a classifier can identify their text with high confidence.

### 1.1 Research Questions

This project addresses the following research questions:

- Can a machine learning model reliably predict which subreddit a piece of text originated from, and which communities are most linguistically distinct?
- What topics naturally emerge within each community, and do they align with the community's stated purpose?
- How does the same word carry different semantic meaning across communities - and can this be quantified?
- Are there temporal patterns in posting behaviour that reveal community-specific rhythms?

### 1.2 Contributions

The primary contributions of this work are:

- A curated dataset of 1,500 posts and 30,000 comments across 10 semantically contrasting subreddits, collected via Reddit's public JSON endpoint without OAuth authentication.
- A multi-model NLP pipeline combining TF-IDF classification, BERTopic topic modelling, and per-community Word2Vec embeddings.
- An original cross-community semantic comparison framework that visualises how identical words carry different meanings in different communities.
- A deployed fullstack web application enabling interactive exploration of all findings.

## 2 Data Collection

### 2.1 Platform and Scope

Reddit was selected as the data source due to its community-centric structure, the volume of text-based interaction, and the availability of public data. Unlike Twitter or Instagram where content is largely individual, Reddit's subreddit structure creates natural community boundaries that make it ideal for comparative linguistic analysis. Each subreddit has its own norms, moderation rules, and implicit expectations about what constitutes a valid post - all of which shape the language used.

The ten selected subreddits are as follows:

- **r/mumbai** – Discussions about Mumbai city, local news, culture, and daily life
- **r/JEENEETards**– Preparation, stress, and memes around JEE/NEET competitive exams
- **r/Btechtards**– Engineering student life, academics, internships, and college culture
- **r/gaming**– Video games, industry news, and gamer discussions

- **r/Fitness**– Exercise routines, fitness advice, and physical health
- **r/depression**– Support community for people dealing with depression
- **r/cats**– Sharing cat photos, stories, and pet-related discussions
- **r/AskReddit**– Open-ended questions and diverse public opinions
- **r/RelationshipAdvice**– Guidance on personal relationships
- **r/AITAH**– Moral judgement and interpersonal conflict

### 2.2 Collection Methodology

Data collection did not use the official Reddit API, which has been significantly restricted since 2023 and requires application approval for meaningful data access. Instead, this project exploits Reddit's lesser-known public JSON endpoint - any Reddit URL can be suffixed with '.json' to receive the page's full data as structured JSON, with no authentication required. For example:

https://www.reddit.com/r/depression/top.json?t=year&limit=100

This approach returns the same data as the official API endpoint but requires no OAuth credentials. A custom Python script using the requests library was developed to paginate through multiple time filters (year, all, month) to collect a diverse temporal sample, with a 2-second delay between requests to avoid rate limiting.

### 2.3 Ethical Considerations

All data collected is publicly accessible - no private subreddits or authenticated content was accessed. Reddit's public JSON endpoint is widely used in academic research. No personally identifiable information beyond usernames (which are pseudonymous by design) was retained in the final dataset. User identifiers were discarded during cleaning.

## 3 Data Cleaning and Preparation

### 3.1 Overview

Raw data collected from Reddit required significant cleaning before it could be used for analysis or modelling. Reddit posts frequently contain deleted content, URLs, and in this dataset's case, significant Hinglish text that standard NLP pipelines are not designed to handle. Cleaning was performed in a dedicated script (cleaner.py) that operates on the raw CSV files and produces cleaned outputs, following the principle of keeping raw data untouched.

### 3.2 Cleaning Steps

- Duplicate removal: Posts and comments were deduplicated using post_id and comment_id respectively to eliminate overlap and repeated entries.
- Junk text handling: Entries with body text marked as [deleted], [removed], or containing only a bare URL were identified. For posts, the title was retained as the primary text in such cases; for comments, these entries were removed.
- Text preparation: For posts, the title and body were concatenated into a single text field to enhance contextual richness. Comments used their body text directly.
- Short text removal: Entries with fewer than 10 characters after processing were discarded due to insufficient information for analysis.
- Whitespace normalisation: Newlines were replaced with spaces, and multiple consecutive spaces were reduced to a single space.
- Datetime processing: The created_utc Unix timestamp was converted into a datetime format, and an hour feature was extracted to support temporal analysis.

### 3.3 Text Preprocessing for NLP

For NLP tasks specifically, an additional preprocessing step was applied beyond the basic cleaning:

- Lowercasing: All text converted to lowercase.
- Tokenisation: NLTK's word_tokenize function was used rather than simple whitespace splitting, as it correctly handles contractions and punctuation.
- Stopword removal: NLTK's English stopword list (180 words) was extended with Reddit-specific terms (reddit, www, amp, gt), a custom Hinglish stopword list of 42 romanised Hindi terms (hai, nahi, bhai, toh, etc.), and a filler word list of semantically empty high-frequency verbs (get, make, think, feel, etc.).
- Lemmatisation: NLTK's WordNetLemmatizer was applied to reduce inflected forms to their base form (running -> run, cats -> cat).
- Short token removal: Tokens under 3 characters were discarded.

After cleaning, approximately 85-90% of rows were retained. The primary source of data loss was short posts (common in r/cats and r/AskReddit where posts are often just image links or single-sentence questions) and deleted content. The final cleaned dataset comprised approximately 1,420 posts and 26,000 comments.

## 4 Exploratory Data Analysis

### 4.1 Overview

Exploratory data analysis was conducted using a dedicated script (eda.py) producing 14 visualisations across five analytical categories: text and vocabulary, engagement, temporal patterns, NLP-derived metrics, and length distributions. All analyses were performed on the combined posts and comments dataset unless otherwise specified.

### 4.2 Text Length Analysis

Post text length varied dramatically across subreddits, revealing fundamental differences in posting culture. r/relationshipAdvice and r/AITAH had the longest posts on average, consistent with their narrative format - users are expected to provide detailed context for their situations before seeking judgement. r/cats and r/gaming had the shortest posts, consistent with their image-centric or reaction-based posting norms.

The lexical diversity analysis revealed an apparent paradox: subreddits with the most unique words per post (r/relationshipAdvice, r/AITAH) had the lowest lexical diversity scores (unique words / total words), while short-post subreddits (r/AskReddit, r/gaming) had the highest diversity scores. This is explained by the nature of the metric: long posts necessarily repeat words more, dragging down the unique/total ratio, while a short 20-word post where 18 words are distinct achieves a near-perfect diversity score regardless of vocabulary richness.

### 4.3 Vocabulary Analysis

The most unique words analysis - identifying words that appear disproportionately in one subreddit relative to others - produced highly interpretable results:

- r/Fitness: gym, squat, bench, protein, rep - pure domain vocabulary with essentially no overlap with other subreddits
- r/JEENEETards: jee, neet, rank, percentile, coaching - Indian competitive exam terminology
- r/cats: kitty, kitten, vet, paw, meow - consistent feline vocabulary
- r/depression: depressed, suicidal, medication, antidepressant, therapist - clinical and emotional vocabulary
- r/mumbai: bmc, marathi, andheri, wankhede - hyperlocal Mumbai references

The subreddit similarity heatmap (TF-IDF cosine similarity between aggregated subreddit corpora) revealed that r/JEENEETards and r/Btechtards are the most similar pair of subreddits in the dataset, reflecting their shared origin in Indian engineering education culture. r/Fitness was the most distinct, with the lowest average similarity to all other subreddits - consistent with its high classifier F1 score.

### 4.4 Engagement Analysis

Score distributions varied significantly across subreddits. r/Fitness showed a wide distribution with frequent high-scoring posts, consistent with its culture of celebrating personal achievements ('progress pics'). r/AskReddit showed the highest volume of very high-scoring posts, reflecting its broad appeal and large subscriber base. The score vs. text length scatter plot showed no consistent linear relationship, suggesting that text length alone is not a predictor of upvote success - content relevance and emotional resonance are likely more important factors.

## 5 Data Visualisation & Dashboarding

### 5.1 Static Visualisations

Static visualisations were produced using Matplotlib and Seaborn as part of the EDA pipeline. These include word clouds, frequency bar charts, KDE plots of text lengths, boxplots of score distributions, heatmaps, line plots of hourly activity, and bar charts of the night owl index. Word clouds and the subreddit similarity heatmap are highlighted here as particularly informative:

Word clouds provided an immediate visual summary of each community's dominant vocabulary. The contrast between subreddits was visually striking — r/Fitness word clouds were dominated by physical training terms, r/depression clouds centred on emotional and clinical language, and r/mumbai clouds featured a mix of English and transliterated Marathi terms.

The TF-IDF cosine similarity heatmap provided a compact summary of inter-community linguistic relationships. The clustering of r/JEENEETards and r/Btechtards, and the isolation of r/Fitness, were immediately apparent from the heatmap's colour structure without requiring any additional interpretation.

### 5.2 Interactive Web Dashboard

An interactive fullstack web application was developed and deployed to enable dynamic exploration of the project's findings. The application uses a React frontend (deployed on Vercel) and a FastAPI backend (deployed on Render), with all ML models served from the backend. The application consists of four pages:

- Overview - Project summary, pipeline diagram, key statistics (total documents, classifier accuracy, topics discovered, vocabulary size), subreddit cards, and key findings
- Classifier - Interactive text input where users can type any text and receive a real-time subreddit prediction with animated confidence bars for all 10 classes. Pre-loaded example texts demonstrate model behaviour across different subreddits.
- Word Explorer - Users can type any word or select from presets (work, money, help, life, body, college, friend) to see the top 7 semantically similar words in each subreddit's embedding space, alongside the pre-computed cosine similarity heatmap for that word.
- EDA Dashboard - All static visualisations organised into categories (Text & Vocabulary, Engagement, Temporal, NLP / ML, Length) accessible via a sidebar navigation, fetched dynamically from the backend's static file server.

Link to the dashboard: *https://reddit-nlp.vercel.app/*

### 5.3 Technical Architecture

The backend serves ML inference via POST endpoints (/classify, /word-explorer) and serves all plot images as static files via a mounted /plots route. This architecture means the frontend never embeds images directly - it always fetches them from the API, demonstrating a clean client-server separation. The VITE_API_URL environment variable enables seamless switching between local development and production endpoints.

## 6 Model Building

### 6.1 Combined Dataset

For all modelling tasks, posts and comments were combined into a single dataset with two columns: text (the post text or comment body) and subreddit (the target label). This decision was made to maximise the volume of training data and to ensure that the models learn from both long-form posts and shorter conversational comments. The combined dataset comprised approximately 27,500 documents across 10 subreddit classes, with roughly balanced class sizes.

## 6.2 Subreddit Classifier

A multi-class text classifier was built to predict the subreddit of origin given a piece of text. The pipeline consists of two components: a TF-IDF vectoriser that converts text to a numerical matrix, and a Logistic Regression classifier trained on that matrix.

TF-IDF (Term Frequency–Inverse Document Frequency) represents each document as a weighted vector of its terms, increasing weights for frequent terms in a document while reducing weights for common terms across the corpus. This highlights distinctive, community-specific vocabulary. The vectoriser used max_features= 10,000 and ngram_range=(1,2) to capture both single words and two-word phrases. Logistic Regression was chosen for its interpretability, efficiency, and strong performance on high-dimensional sparse text data. The model was trained with max_iter=1000 and C=1.0, using a one-vs-rest approach to handle the 10-class classification problem and generate class probability estimates for the web application.

Overall accuracy of 64.13% represents a substantial improvement over the random baseline of 10% for a 10-class problem. r/Fitness achieved the highest F1 score (0.83), reflecting its highly domain-specific vocabulary with minimal overlap with other subreddits. r/mumbai and r/Btechtards scored lowest (0.55, 0.53), primarily due to Hinglish content that the model struggles to distinguish from r/JEENEETards - both subreddits share similar cultural vocabulary. r/AskReddit and r/gaming are frequently confused with each other, as both feature casual conversational text without strong domain-specific markers.

## 6.3 Topic Modelling - BERTopic

BERTopic was used for topic modelling, selected over traditional LDA due to its use of transformer-based sentence embeddings that capture semantic meaning rather than relying purely on word co-occurrence. BERTopic clusters documents using HDBSCAN and represents topics using class-based TF-IDF (c-TF-IDF), producing interpretable topic labels from the most distinctive terms in each cluster.

BERTopic discovered 18 coherent topics from the combined corpus. Topic -1 (the outlier/noise bucket) contained 13,425 documents that the algorithm could not confidently assign, which is typical for conversational text. Notable topics included:

- Topic 0 (1,880 docs): gym, weight, set, bench, workout, squat - clearly r/Fitness
- Topic 1 (993 docs): cat, kitty, kitten, vet, litter - clearly r/cats
- Topic 2 (641 docs): bhai, nhi, kar, kuch - Hinglish cluster spanning r/Btechtards and r/JEENEETards
- Topic 6 (266 docs): depression, depressed, medication, antidepressant - r/depression clinical language
- Topic 5 (268 docs): trump, republican, president, american - political discussion in r/AskReddit

The Hinglish cluster (Topic 2) is a particularly significant finding - it emerged entirely without manual intervention, demonstrating that the model naturally identified Hindi as a distinct linguistic register. This cluster cuts across subreddit boundaries, appearing in both r/Btechtards and r/JEENEETards, suggesting that Hinglish is a community-level phenomenon rather than a subreddit-specific one.

## 6.4 Word Embeddings - Word2Vec

Word2Vec models were trained separately for each subreddit to capture community-specific semantic associations. The skip-gram architecture was used with vector_size=100, window=5, min_count=3, and 10 training epochs. Training a separate model per subreddit, rather than a single global model, allows direct comparison of how the same word is positioned differently in each community's semantic space.

The core finding of the word embedding analysis is that identical words carry fundamentally different semantic associations across communities. Selected findings:

- 'life' in r/depression - suck, sad, worth, either (negative emotional framing); in r/relationshipAdvice - live, move, deserve, future (forward-looking framing); in r/Fitness - consistently, consistent, hiatus (performance framing)
- 'help' in r/relationshipAdvice - therapy, therapist, professional (institutional support); in r/JEENEETards - bro, study, education (peer support in academic context)
- 'body' in r/Fitness - muscle, lose, loss, carb (physical composition); in r/gaming - gun, hand, control (game mechanics)
- 'college' in r/JEENEETards - exam, rank, teacher (entrance exam focus); in r/Btechtards - internship, tier, placement (career focus)

# 7 Final Remarks

What this project ultimately demonstrates is that language is not just a tool for communication - it is a marker of identity. The communities on Reddit have not just chosen what to talk about; they have collectively built distinct ways of talking. The methods applied here - from TF-IDF to Word2Vec to BERTopic - are not merely classification tools. They are lenses through which the linguistic fingerprint of each community becomes visible. In that sense, this project is as much about understanding communities as it is about building models. Link to the GitHub repository: *https://github.com/siddhisingh9/RedditNLP*